EFY TESTED
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OBJECT DETECTION And CLASSIFICATION Using Edge Impulse On Raspberry Pi

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If machines could recognise objects the way human beings do, it would be quite interesting. Object detection is a trending topic nowadays. So let us make an object detection camera that can do live classification of objects and people, and every activity is displayed live on web using an IP address.

We will use Edge Impulse to train an ML model and integrate it with a computing machine (Raspberry Pi) and acquire live video and images through camera interfacing. The project will be able to work with the internet and without the intranet. Using the internet, we can use the project for activities like:

- Live door monitoring to alert when an unknown person enters
- In industry to do object classification and separation using robotic arms
- Counting of fruits on a tree or in a separator

There are various versions and flavours of OS available for Raspberry Pi but we need to prepare the SD card with the latest OS. To load the OS on SD card, follow the steps mentioned below:

1. Download Raspberry Pi Desktop Imager on a PC.
2. Launch the Raspberry Pi Imager
3. Choose OS as Raspberry Pi OS (32-bit)
4. Choose SD card
5. Select Write
6. Insert SD card into Raspberry Pi
7. Connect Raspberry Pi to power, keyboard, mouse, and monitor
8. If OS is properly installed, you will see "Welcome to Raspberry Pi Desktop" message.

Now, with the OS ready, connect the USB camera or Raspberry Pi camera. If Raspberry Pi camera is used (connected with a ribbon

BILL OF MATERIAL

1. Raspberry Pi 3 B-1
2. USB camera/Raspberry Pi camera-1
3. Keyboard-1
4. Monitor-1
5. Mouse-1
7. SD Adaptor (32GB)
8. HDMI to VGA cable
9. 5V power adaptor with USB Type-C connector
10. SD card reader

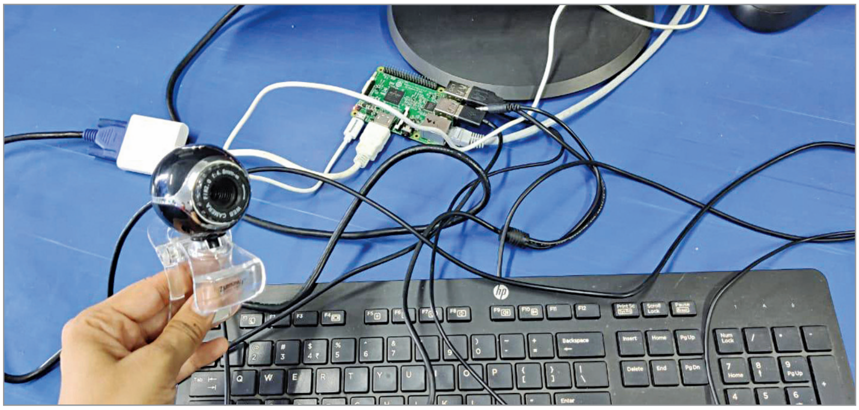

Fig. 1: Author's prototype

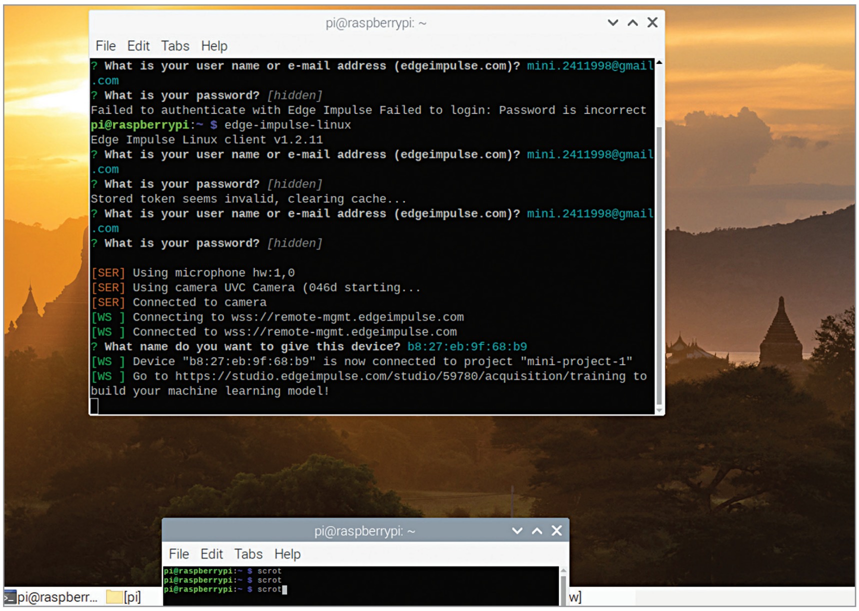

Fig. 2: Running Edge Impulse engine